

Bill of Materials - VBF-T2R2-BOM-01

Case t2_r2 - Virtual Battery Factory - generated 2026-08-26

Bill of Materials (dual caliber: g/cell and kg/kWh; formulas annotated; missing items N-A)

Component	Formula (mass)	g / cell	kg / kWh	Annotation
Positive electrode active material (NMC)	coating t x area x AMVF x rho (eq. solids)	24.8927	0.9819	
Positive electrode conductive additive	literature default 1.5 wt% of coating solids	0.3869	0.0153	annotation: no CA parameter in set
Positive electrode binder	literature default 2.0 wt% of coating solids	0.5159	0.0204	annotation: no binder parameter in set
Negative electrode active material (graphite)	coating t x area x AMVF x rho (eq. solids)	14.2869	0.5636	
Negative electrode conductive additive	literature default 1.5 wt% of coating solids	0.2221	0.0088	annotation: no CA parameter in set
Negative electrode binder	literature default 2.0 wt% of coating solids	0.2961	0.0117	annotation: no binder parameter in set
Separator	calc layer_kg_m2 x area	0.1835	0.0072	
Electrolyte	pore volume x electrolyte density	12.5089	0.4934	annotation: electrolyte density = literature 1.2 g/cm3; fill factor 1.0
Positive current collector (Al)	calc layer_kg_m2 x area	2.2183	0.0875	
Negative current collector (Cu)	calc layer_kg_m2 x area	5.5212	0.2178	
Enclosure (can/pouch)	Not modeled	-	-	N-A (no parameter)
Tabs / terminals	Not modeled	-	-	N-A (no parameter)
SUMMARY - total (contract caliber)	= calc mass_kg x1000	48.5236	1.9141	electrolyte excluded: ■■■■■■■■■■■■■■■■■■■■■
SUMMARY - total incl. electrolyte	annotation extension	61.0325	2.4075	electrolyte density 1.2 g/cm3 literature
SUMMARY - total energy	final_calc.json	25.3513 Wh (= 0.025351 kWh)		